

Tabel cu integralele funcțiilor compuse

	Formula	Exemplul
1.	$\int u^n \cdot u' = \frac{u^{n+1}}{n+1} + C$	$\int (x^2 + 1)^9 \cdot 2x \, dx = \frac{(x^2 + 1)^{10}}{10} + C$
2.	$\int u \cdot u' = \frac{u^2}{2} + C$	$\int \ln x \cdot \frac{1}{x} \, dx = \frac{\ln^2 x}{2} + C$
3.	$\int \sqrt{u} \cdot u' = \frac{2}{3} \cdot u \cdot \sqrt{u} + C$	$\int \sqrt{\sin x} \cdot \cos x \, dx = \frac{2}{3} \cdot \sin x \cdot \sqrt{\sin x} + C$
4.	$\int \frac{1}{u} \cdot u' = \ln u + C$	$\int \frac{1}{x^2 + 1} \cdot 2x \, dx = \ln(x^2 + 1) + C$
5.	$\int e^u \cdot u' = e^u + C$	$\int e^{x^2} \cdot 2x \, dx = e^{x^2} + C$
6.	$\int a^u \cdot u' = \frac{a^u}{\ln a} + C$	$\int 2^{\sin x} \cdot \cos x \, dx = \frac{2^{\sin x}}{\ln 2} + C$
7.	$\int \sin u \cdot u' = -\cos u + C$	$\int \sin(x^3 + 3x) \cdot (3x^2 + 3) \, dx = -\cos(x^3 + 3x) + C$
8.	$\int \cos u \cdot u' = \sin u + C$	$\int \cos x^3 \cdot x^2 \, dx = \frac{1}{3} \cdot \sin x^3 + C$
9.	$\int \operatorname{tg} u \cdot u' = -\ln \cos u + C$	$\int e^x \cdot \operatorname{tg} e^x \, dx = -\ln \cos e^x + C$
10.	$\int \operatorname{ctg} u \cdot u' = \ln \sin u + C$	$\int \operatorname{ctg} \sqrt{x} \cdot \frac{1}{\sqrt{x}} \, dx = 2 \cdot \ln \sin \sqrt{x} + C$
11.	$\int \frac{1}{\sin^2 u} \cdot u' = -\operatorname{ctg} u + C$	$\int \frac{1}{\sin^2 x^2} \cdot x \, dx = -\frac{1}{2} \cdot \operatorname{ctg} x^2 + C$
12.	$\int \frac{1}{\cos^2 u} \cdot u' = \operatorname{tg} u + C$	$\int \frac{1}{\cos^2 (3x + 1)} \, dx = \frac{1}{3} \cdot \operatorname{tg} (3x + 1) + C$
13.	$\int \frac{u'}{u^2 - a^2} = \frac{1}{2a} \cdot \ln \left \frac{u - a}{u + a} \right + C$	$\int \frac{1}{x(\ln^2 x - 3^2)} \, dx = \frac{1}{2 \cdot 3} \cdot \ln \left \frac{\ln x - 3}{\ln x + 3} \right + C$
14.	$\int \frac{u'}{u^2 + a^2} = \frac{1}{a} \cdot \operatorname{arctg} \frac{u}{a} + C$	$\int \frac{3x^2}{(x^3)^2 + 7^2} \, dx = \frac{1}{7} \cdot \operatorname{arctg} \frac{x^3}{7} + C$
15.	$\int \frac{u'}{\sqrt{u^2 + a^2}} = \ln(u + \sqrt{u^2 + a^2}) + C$	$\int \frac{e^x}{\sqrt{e^{2x} + 4}} \, dx = \ln(e^x + \sqrt{e^{2x} + 4}) + C$
16.	$\int \frac{u'}{\sqrt{u^2 - a^2}} = \ln \left u + \sqrt{u^2 - a^2} \right + C$	$\int \frac{2x}{\sqrt{(x^2)^2 - 1}} \, dx = \ln \left x^2 + \sqrt{(x^2)^2 - 1} \right + C$
17.	$\int \frac{u'}{\sqrt{a^2 - u^2}} = \operatorname{arcsin} \frac{u}{a} + C$	$\int \frac{x^3}{\sqrt{1 - x^8}} \, dx = \frac{1}{4} \cdot \operatorname{arcsin} x^4 + C$

$$\overset{\text{notație}}{u(x)} \cong u \quad \text{și} \quad \overset{\text{notație}}{u'(x)dx} \cong u'$$